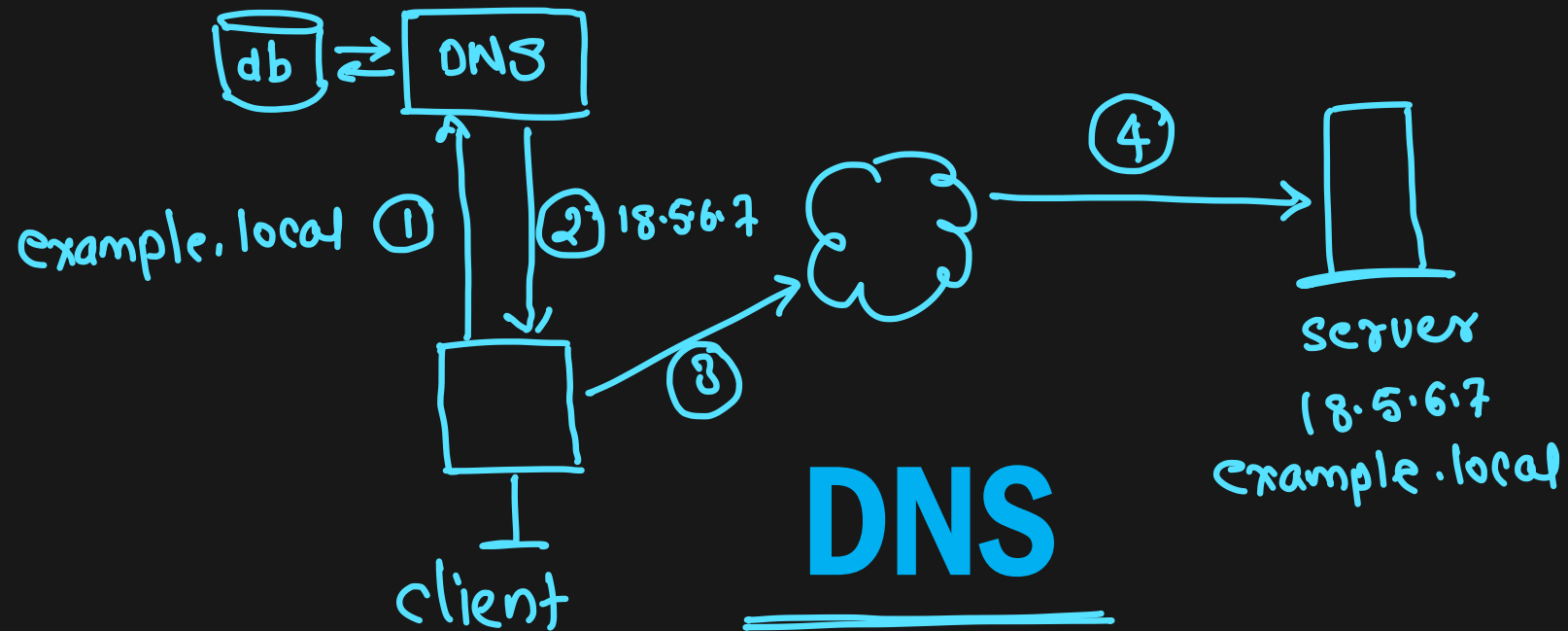




DNS

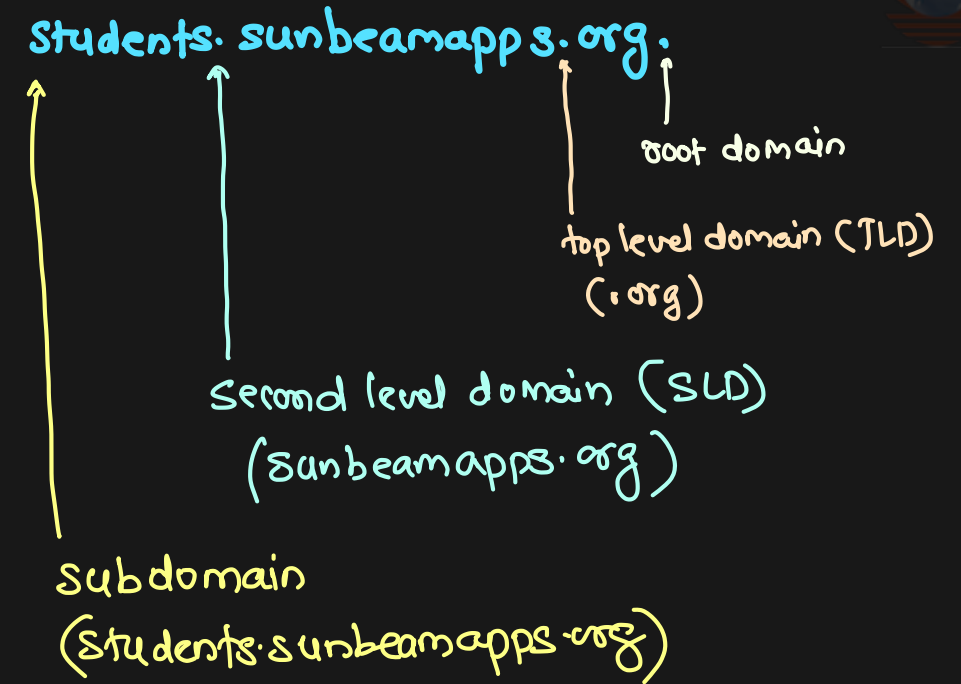


- The Domain Name System (DNS) is a hierarchical and distributed naming system for computers, services, and other resources in the Internet or other Internet Protocol (IP) networks
- It associates various information with domain names assigned to each of the associated entities
- Most prominently, it translates readily memorized domain names to the numerical IP addresses needed for locating and identifying computer services and devices with the underlying network protocols
- The Domain Name System has been an essential component of the functionality of the Internet since 1985
- Uses port no 53

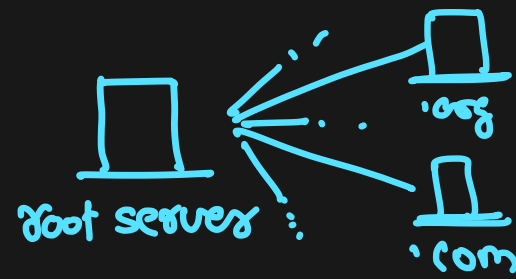
local DNS → /etc/hosts

Hierarchy of Domain Names

- Root Domain:
 - The dot (.) at the end (often invisible). Example: example.com.
- Top-Level Domain (TLD):
 - .com, .org, .net, .uk, .edu
- Second-Level Domain (SLD)
 - The registrable part. Example: google in google.com
- Subdomain
 - Optional prefix. Example: mail.google.com (mail is subdomain)



Terminologies



■ DNS Root Servers

- Name server for root zone of DNS of the internet
- It directly answers requests for records in the root zone and answers other requests by returning a list of the authoritative name servers for the appropriate Top Level Domain (TLD)
- TLDs are controlled by the root zone

■ Top Level Domain (TLD)

- A top-level domain (TLD) is one of the domains at the highest level in the hierarchical Domain Name System of the Internet
- IANA currently distinguishes the following groups of top-level domains:
 - country-code top-level domains (ccTLD) → .in, .us, .uk
 - generic top-level domains (gTLD)
 - sponsored top-level domains (sTLD) - .com
 - unsponsored top-level domains . edu, .org
 - infrastructure top-level domain (.arpa)

Terminologies



■ Subdomain

- Anything between the host and TLD is known as subdomain
- E.g. sunbeaminfo.com is subdomain of .com TLD

■ Zone and zone files → database ↴

- A zone of zone file is a mapping of IP address and the hosts for a given subdomain
- The zone file for sunbeaminfo.com would contain a record for www

■ Name Server

- A server that runs the DNS service and can either store or cache the information for the DNS platform
- Whether it stores or caches depends on whether it is referred from a higher level

■ Authoritative

- The organizations run name servers that become authoritative
- They can answer the queries with authority

Terminologies



machine / printer / . .

■ Resource Record (RR)

- This is information of element for network resources
- The types of information elements are categorized and organized with a list of DNS record types
- Resource records of the same type are described as a *resource record set* (RRset)
- DNS resolvers return the entire set upon query, but servers may implement round-robin ordering to achieve load balancing
- Each record has a
 - NAME: Name of the node to which this record pertains
 - TYPE: Type of RR in numeric form (e.g., 15 for MX RRs)
 - CLASS: Class code
 - TTL: Count of seconds that the RR stays valid (The maximum is $2^{31}-1$, which is about 68 years)
 - RDATA: Additional RR-specific data
 - RDLENGTH: Length of RDATA field (specified in octets)

Terminologies



■ Hosts

- A record in the zone file: www

■ FQDN

- Fully Qualified Domain Name
- The host and the domain name

■ Registrar

- A domain name registrar is a company that manages the reservation of Internet domain names
- A domain name registrar must be accredited by a generic top-level domain (gTLD) registry or a country code top-level domain (ccTLD) registry
- A registrar operates in accordance with the guidelines of the designated domain name registries

■ Resolver

- The client side of the DNS is called a DNS resolver
- A resolver is responsible for initiating and sequencing the queries that ultimately lead to a full resolution (translation) of the resource sought, e.g., translation of a domain name into an IP address

Types of DNS Records → RR



- ✓ **A (Address):** Maps domain to IPv4 address
- ✓ **AAAA:** Maps domain to IPv6 address
- **CNAME (Canonical Name):** Aliases one domain to another (e.g., www to @)
- **MX (Mail Exchange):** Directs email to mail servers (with priority values)
- **TXT (Text):** Holds human/machine-readable text (used for verification)
- **NS (Name Server):** Specifies the authoritative DNS servers for the domain
- **PTR (Pointer):** Maps IP to domain (reverse DNS lookup)
- **SOA (Start of Authority):** Contains administrative info about the zone



Types of DNS Servers

- **Recursive Resolver:** Does all the legwork (steps 3-6) for the client
- **Root Name Server:** Top of hierarchy (13 logical sets worldwide). Knows TLD servers
- **TLD Name Server:** Handles specific extension (.com, .org, etc.). Knows Authoritative servers
- **Authoritative Name Server:** Final source of truth for a domain's records

How DNS Resolution Works (8 Steps)



1. **User types** example.com into browser
2. **Check cache:** Browser to OS to Router to ISP Resolver (if found, skip to end)
3. **Query Root Server:** Resolver asks root "Where is .com?" Root replies with TLD server address,
4. **Query TLD Server:** Resolver asks .com server "Where is example.com?" TLD replies with Authoritative server
5. **Query Authoritative Server:** Resolver asks the domain's own DNS server "What is the IP of example.com?"
6. **Response:** Authoritative server returns A record (IP address)
7. **Caching:** Resolver caches IP for future queries (based on TTL)
8. **Deliver:** Resolver sends IP to browser, browser loads website

